Computer Networking: Security (CSE-3752)
Experiment-5A
DES Round Key Generation

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Objective-1: Write a Python program to generate 16 round keys (48-bit each) from a 64-bit DES key.

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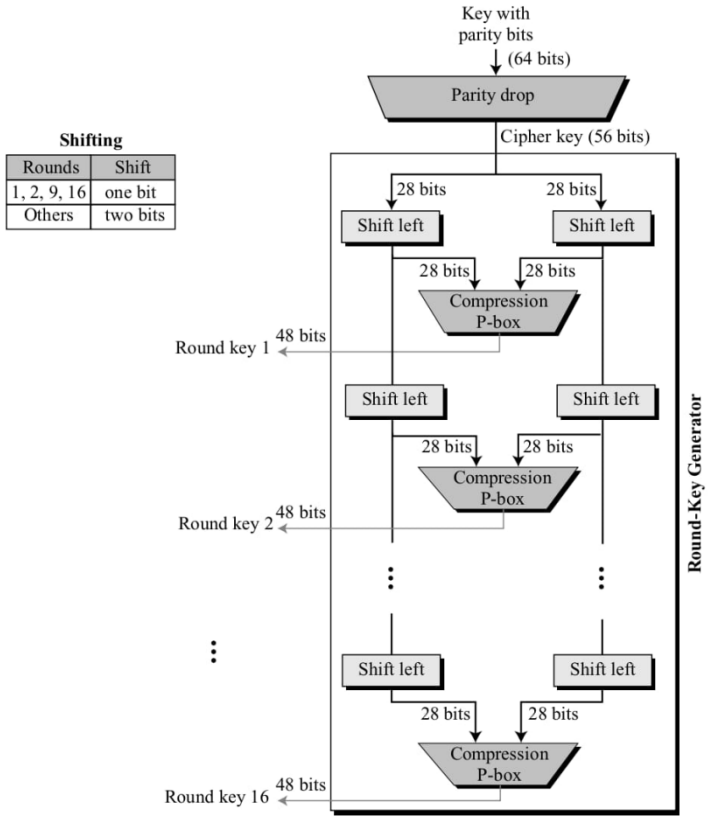
The key generation process is defined in the following figure:



Parity Drop Table:

57	49	41	33	25	17	09	01
58	50	42	34	26	18	10	02
59	51	43	35	27	19	11	03
60	52	44	36	63	55	47	39
31	23	15	07	62	54	46	38
30	22	14	06	61	53	45	37
29	21	13	05	28	20	12	04

Shift Left Table:

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bit shifts	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

Compression P-Box Table:

14	17	11	24	01	05	03	28
15	06	21	10	23	19	12	04
26	08	16	07	27	20	13	02
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Task 1: Define Required Tables

Create the following tables in your code:

- Parity Drop Table
- Shift Left Table
- Compression P-Box Table

Task 2: Write Format Conversion Functions

Write two methods ***hex_to_bin(str)***and ***bin_to_hex(str)***to convert the hexadecimal string to binary and vice versa. Missing bits should be filled with 0s.

Task 3: Write Permutation Function

Write a method***permute(data, table)***to rearrange bits by referring a table.

Task 4: Write Left Shift Function

Write a method ***shift_left(data, n)***to perform circular left shift operation.

Task 5: Write Key Generation Function

Write a method ***generate_round_keys(key_64bit)***to generate 16 round keys.

- Divide into two halves: left and right.
- Perform circular left shift for both the halves by referring the shift left table.
- Merge the results of both halves.
- Apply compression P-Box table to get 48 bit round key in binary.
- Append the round key to ***round_keys*** list and return *round_keys*.

Task 6: Input Handling

- Write code to accept a key in hexadecimal(16 hex digits)and convert it to 64-bit binary.
- Call ***generate_round_keys(key_64bit)*** methodto get***round_keys***list.
- Display the ***round_keys***that are in hexadecimal.